

PROJECT REPORT ON

WAZUH OPEN-SOURCE SIEM TOOL

PROJECT ENVIRONMENT

Operating Systems	Kali Linux & Windows
SIEM Platform	Wazuh
Endpoint Monitoring	Sysmon
Security Framework	MITRE ATT&CK
Command & Scripting	CMD & PowerShell

Submitted By

ADORN CYRIAC MATHEW

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INTRODUCTION

Unauthorized or unexpected modifications to sensitive files, repeated authentication failures, system discovery activities, and attempts to evade security controls can indicate security breaches, malware activity, or other suspicious behavior. Security Information and Event Management (SIEM) solutions play a crucial role in detecting and correlating such activities and providing security teams with timely alerts for investigation.

This project focuses on implementing Wazuh SIEM for security monitoring, threat detection, and file integrity monitoring across Linux and Windows environments. A Wazuh agent was deployed on a Linux virtual machine to continuously monitor predefined sensitive files and directories using real-time monitoring and integrity scans. The Wazuh server processed these events and applied custom alert rules to generate meaningful security alerts.

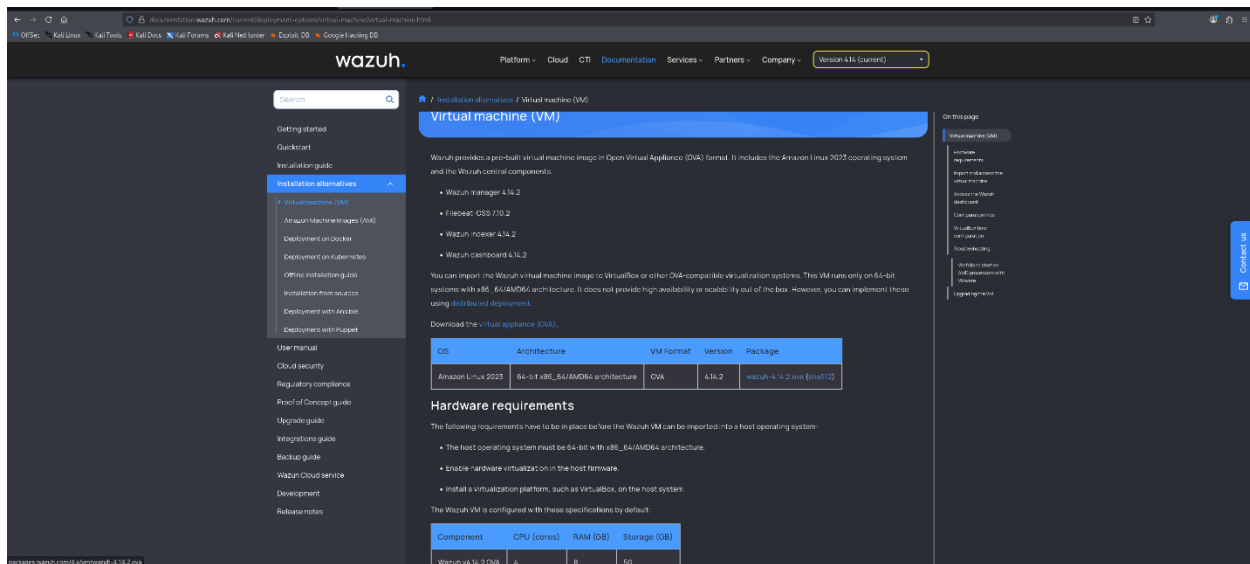
The project also included controlled security-testing activities in a Windows environment using Sysmon and Wazuh to validate detection capabilities. Simulated brute-force attempts, discovery activities, and defense-evasion techniques were executed and monitored through the Wazuh dashboard. The resulting security events and alerts were analyzed to understand the detected activity, affected systems, and relevant security telemetry.

For file integrity monitoring, Wazuh syscheck was configured with real-time monitoring and change reporting to provide timely visibility into modifications to sensitive files. The scan configuration was optimized for faster detection, while a custom server-side alert rule was implemented to escalate relevant file integrity events to a higher severity level and provide details such as the modified file path.

The project demonstrates the practical use of Wazuh SIEM for centralized security monitoring, alert analysis, threat detection, event correlation, and file integrity monitoring. It also provides hands-on experience in analyzing security alerts generated from controlled attack simulations and understanding how SIEM telemetry can support proactive incident detection and response.

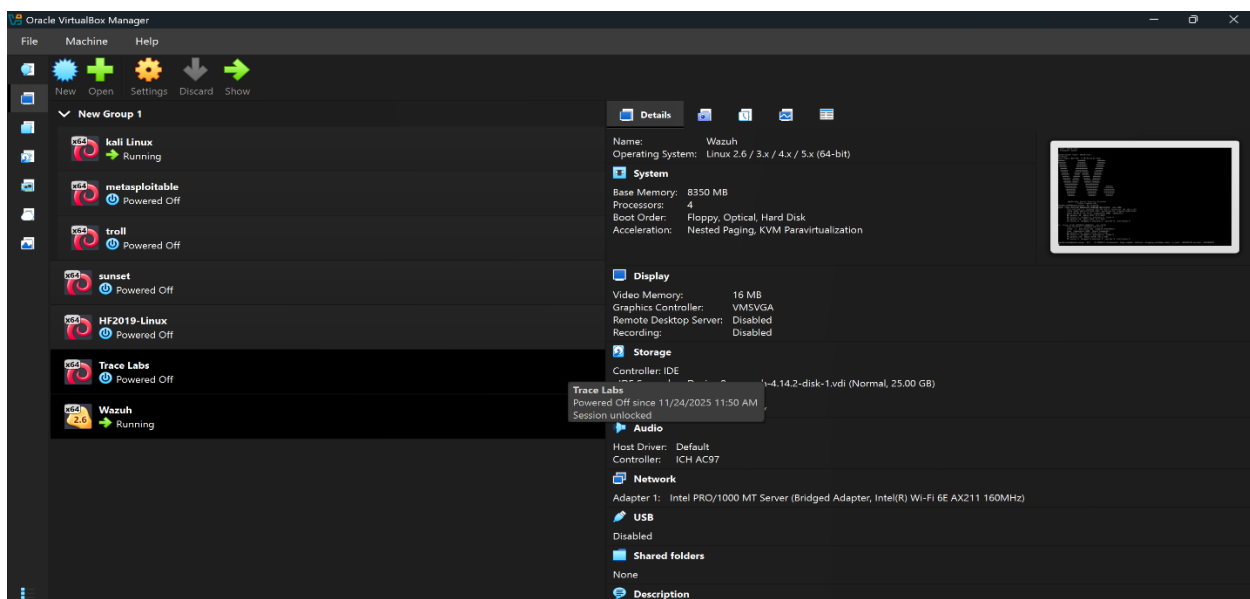
1.Setting up the Wazuh Server

a. Download the Wazuh OVA file (v4.14.2) from the official Wazuh website



b. Import the Wazuh 4.14.2.ova file to your virtualization platform.

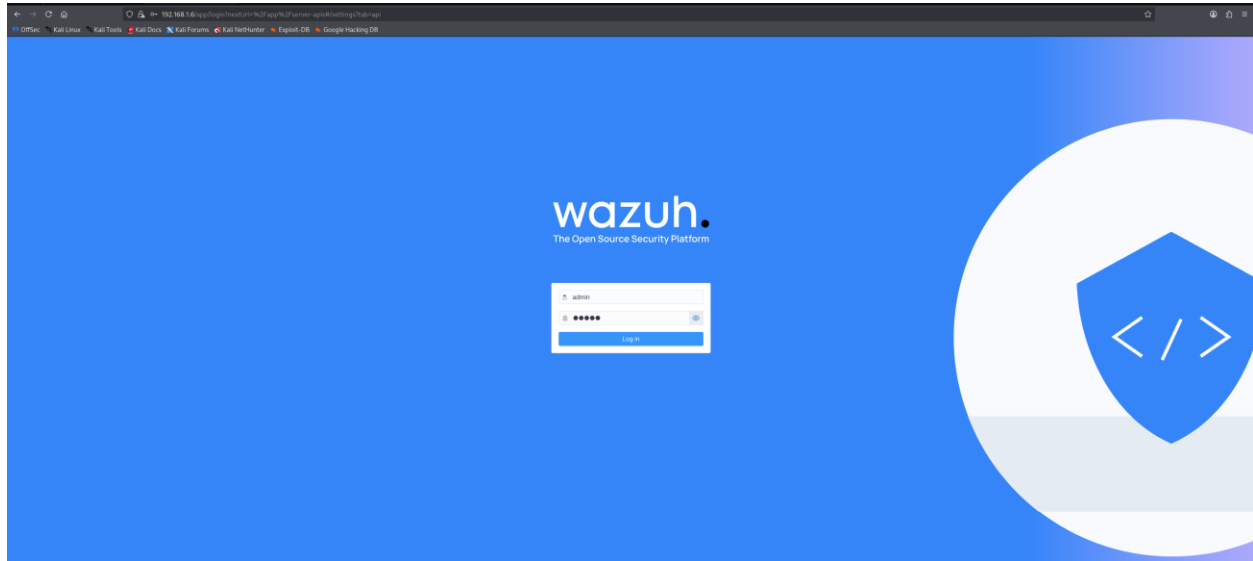
- Select the imported VM settings
- Change the Base Memory to sufficient for smooth functioning
- Change the network to Bridged Adapter
- Change the Graphics Controller to VMSVGA option



3. Accessing the Wazuh Dashboard

After starting the VM, access the Wazuh dashboard in a web browser. Enter the following URL in the address bar

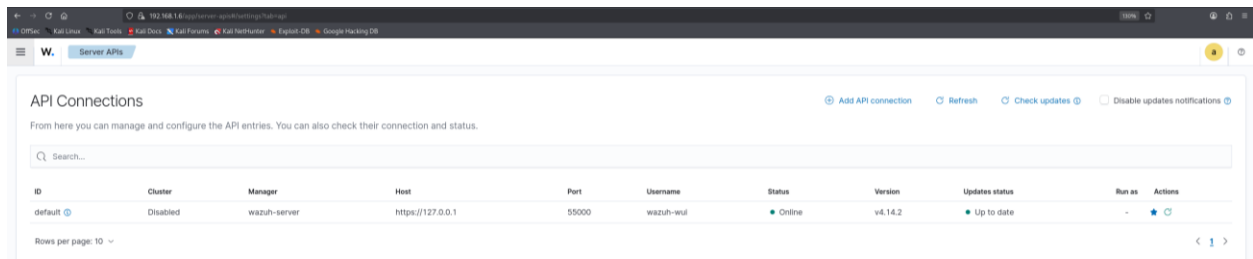
url: https://<WAZUH-IP>

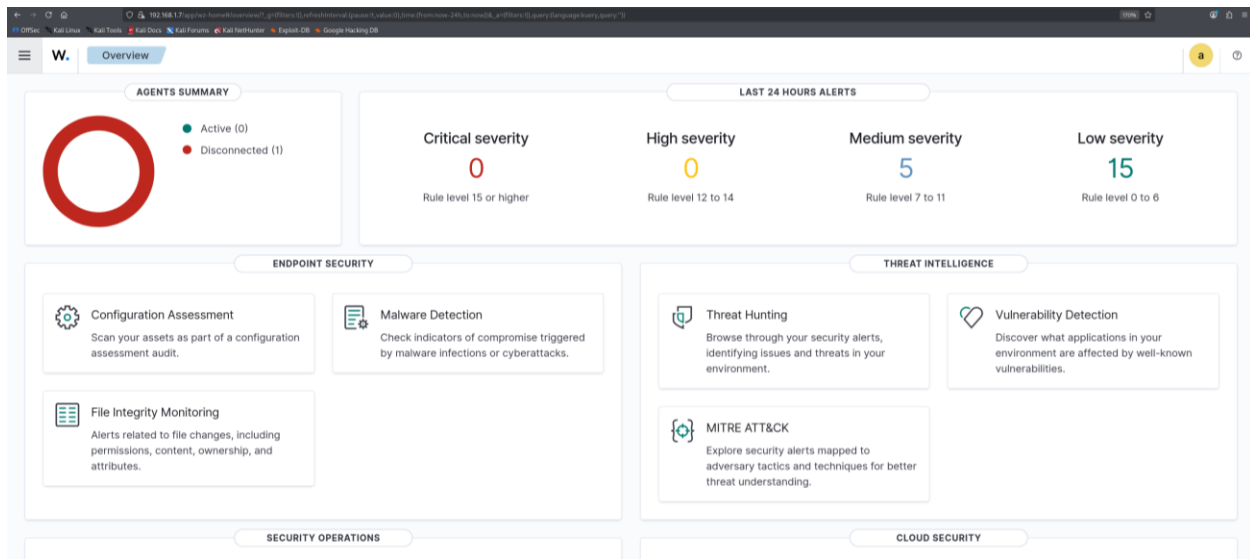


Login using the default credentials

Username: admin

Password: admin





4. Installing Wazuh Agent on Linux

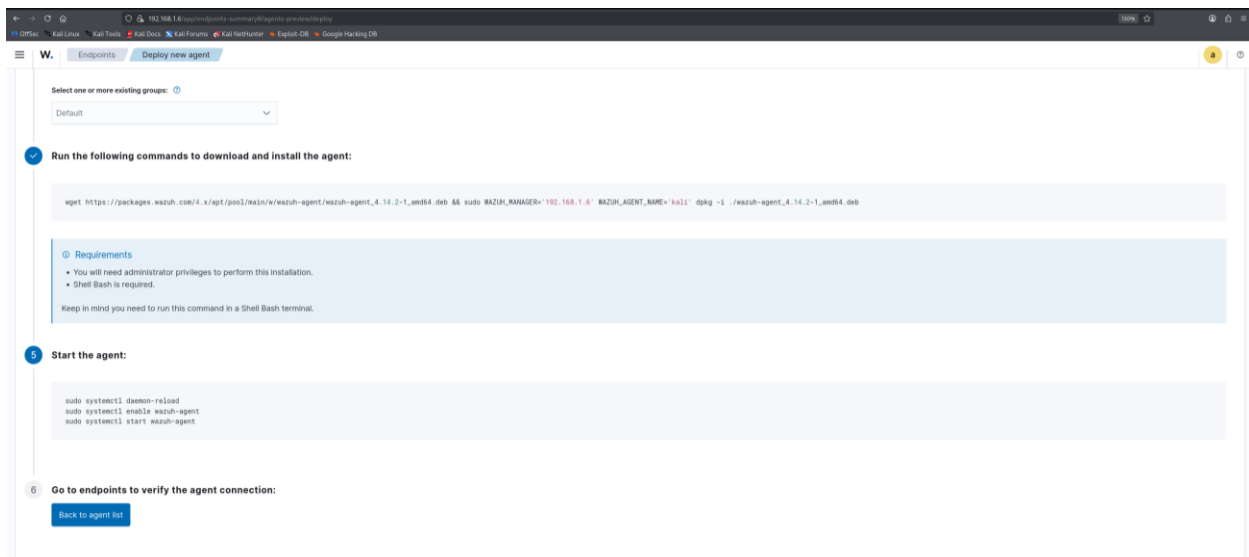
Once the “Deploy new agent” button is clicked from the Wazuh Dashboard /selecting the “Agents Summary”, the deployment configuration window appears

The screenshot shows the 'Deploy new agent' configuration window in the Wazuh Dashboard. It is divided into three main sections. The first section, 'Select the package to download and install on your system:', allows choosing between LINUX, WINDOWS, and macOS. Under the LINUX tab, the 'DEB amd64' option is selected. The second section, 'Server address:', includes a text input field with the IP address '192.168.1.6' and a checked checkbox for 'Remember server address'. The third section, 'Optional settings:', features a text input field for 'Assign an agent name:' with the value 'k-wd'.

Select the Operating System

Select the DEB amd64 bits option to download the installer compatible with Linux system.

Enter the Wazuh Manager IP Address



After filling in all details, the Wazuh Dashboard automatically generate a terminal command to simplify the download and installation process.

- Running this command on linux root terminal

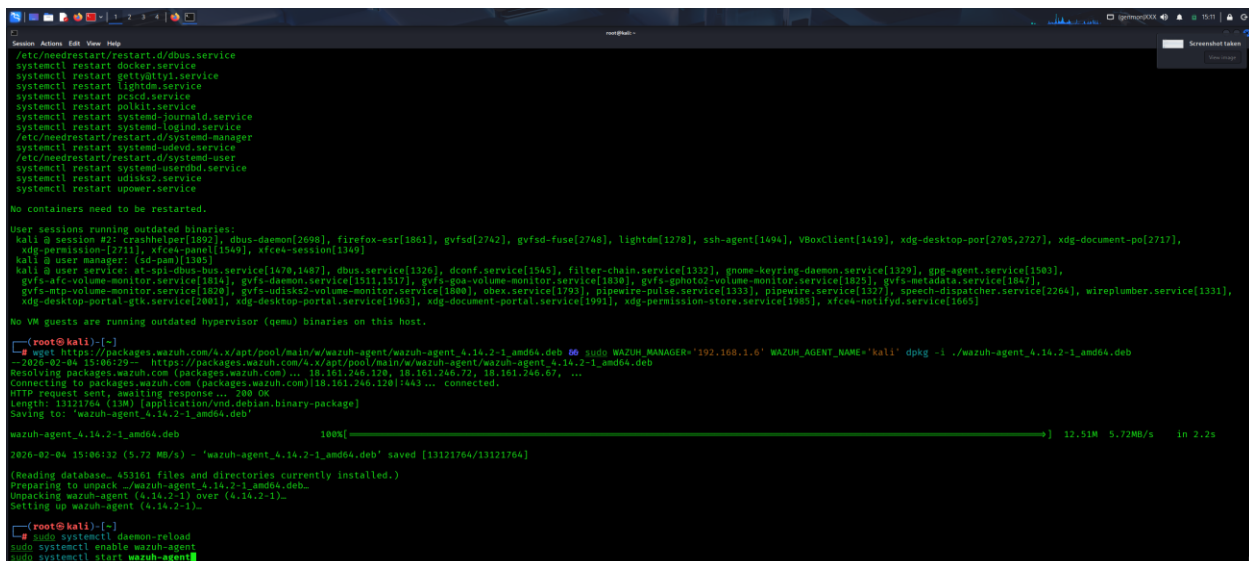
“`wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.2-1_amd64.deb && sudo WAZUH_MANAGER='192.168.1.7' WAZUH_AGENT_NAME='kali' dpkg -I ./wazuh-agent_4.14.2-1_amd64.deb`

- After running the installation command, start the agent by entering the following command

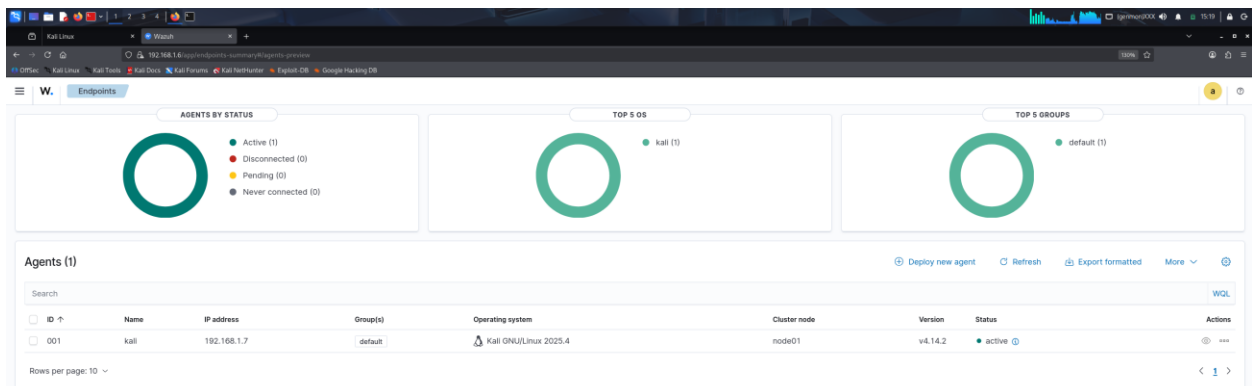
`Sudo systemctl daemon-reload`

`Sudo systemctl enable wazuh-agent`

`Sudo systemctl start wazuh-agent`



To confirm the connection, check the Wazuh dashboard



5. Configuring Monitoring rule in Agent

To configure the rule, follow these steps in linux terminal:

- Type `nano /var/ossec/etc/ossec.conf`

```
zsh: corrupt history file /root/.zsh_history
[root@kali]~# nano /var/ossec/etc/ossec.conf
[root@kali]~# nano /var/ossec/etc/ossec.conf
[root@kali]~#
```

- Declaring server ip address inside `<address></address>` tag for communication.

```
[root@kali]~# nano /var/ossec/etc/ossec.conf
Wazuh - Agent - Default configuration for kali 2025.4
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh

<ossec_config>
  <client>
    <server>
      <address>192.168.1.7</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>kali, kali2025, kali2025.4</config-profile>
    <notify_time>20</notify_time>
    <time-reconnect>30</time-reconnect>
    <auto_restart>yes</auto_restart>
    <crypto_method>aes</crypto_method>
  </client>
```

- Increasing the frequency of checking to 60 by changing 43200, which is the default value.

```

Session Actions Edit View Help
OSSEC Home 8.7
~/ossec/etc/ossec.conf *
</scs>

#----- File integrity monitoring -----
<syscheck>
  <disabled>no</disabled>

  #----- Frequency that syscheck is executed default every 12 hours ---->
  <frequency>12</frequency>

  <scan_on_start>yes</scan_on_start>

  #----- Directories to check (perform all possible verifications) ---->
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>bin/sbin/boot</directories>

  #----- Files/directories to ignore ---->
  <ignore>/etc/mtab</ignore>
  <ignore>/etc/hosts.deny</ignore>
  <ignore>/etc/mail/statistics</ignore>
  <ignore>/etc/random.seed</ignore>
  <ignore>/etc/random.seed</ignore>
  <ignore>/etc/adjtime</ignore>
  <ignore>/etc/httpd/logs</ignore>
  <ignore>/etc/utmpx</ignore>
  <ignore>/etc/wtmpx</ignore>
  <ignore>/etc/cups/certs</ignore>
  <ignore>/etc/dumpdates</ignore>
  <ignore>/etc/svc/volatile</ignore>

  #----- File types to ignore ---->
  <ignore type>"sregex">.log$.swp$</ignore>

  #----- Check the file, but never compute the diff ---->
  <nodiff>/etc/ssl/private.key</nodiff>

  <skip_ifs>yes</skip_ifs>
  <skip_devs>yes</skip_devs>
  <skip_procs>yes</skip_procs>
  <skip_sys>yes</skip_sys>

```

- Declaring the Directories for which requires realtime monitoring.

Type the following command as last line inside `<syscheck>` `</syscheck>` tag :

```
<directories realtime= "yes" report_changes= "yes" check_all=
"yes">/etc./usr/bin./usr/sbin</directories>
```

[illegible]

Save the file and restart the agent

```
Feb 06 03:20:26 kali env[6477]: Started wazuh-syscheckd...
Feb 06 03:20:27 kali env[6477]: Started wazuh-logcollector ...
Feb 06 03:20:28 kali env[6477]: Started wazuh-modulesd...
Feb 06 03:20:30 kali env[6477]: Completed.
Feb 06 03:20:30 kali systemd[1]: Started wazuh-agent.service - Wazuh agent.

[root@kali]:[~]
# nano /Var/ossec/etc/ossec.conf
[root@kali]:[~]
# systemctl restart wazuh-agent
#
```

6. Configuring Alert

After successfully integrating the linux agent, custom alert were configured on the Wazuh server to monitor specific log activity ,file modifying log. These rules enhance system security by monitoring file integrity, generating meaningful alerts, and supporting proactive incident response.

Editing the Local Rules File

```
GNU nano 8.3 /var/ossec/etc/rules/local_rules.xml Modified
<!-- Local rules -->

<!-- Modify it at your will. -->
<!-- Copyright (C) 2015, Wazuh Inc. -->

<!-- Example -->
<group name="local,syslog,sshd,">

  <!--
  Dec 10 01:02:02 host sshd[1234]: Failed none for root from 1.1.1.1 port 1066 ssh2
  -->
  <rule id="100001" level="5">
    <if_sid>5716</if_sid>
    <srcip>1.1.1.1</srcip>
    <description>sshd: authentication failed from IP 1.1.1.1.</description>
    <group>authentication_failed,pci_dss_10.2.4,pci_dss_10.2.5,</group>
  </rule>
</group>

<group name="syscheck,custom_fin">
<rule id="100500" level="14">
<if_sid>550</if_sid> <!--Default file modified SID -->
<description>CRITICAL: A monitored file system file $(file) has been modified</description>
</rule>
</group>
```

Added the following rule in the local rule file:

```
<group name= "custom_file_mod">

<rule id= "100500" level= "14">

<if_sid>550</if_sid>

<description>CRITICAL: A monitored file system $(file) has been modified</description>

</rule>
```

The implemented rules :

- Increased the severity level to “14”.
- <if_sid>550</if_sid> triggers a custom rule when the base rule ID 550 ("Integrity checksum changed") is activated.
- Rule ID 100300 will only fire if rule 550 is triggered.
- Description gives clear indication and providing exact path.

Save the file and restart the Wazuh manager using “systemctl restart wazuh-manager”

7. Verifying the Custom Alert Functionality

1. Performing the test

To validate the alert configuration:

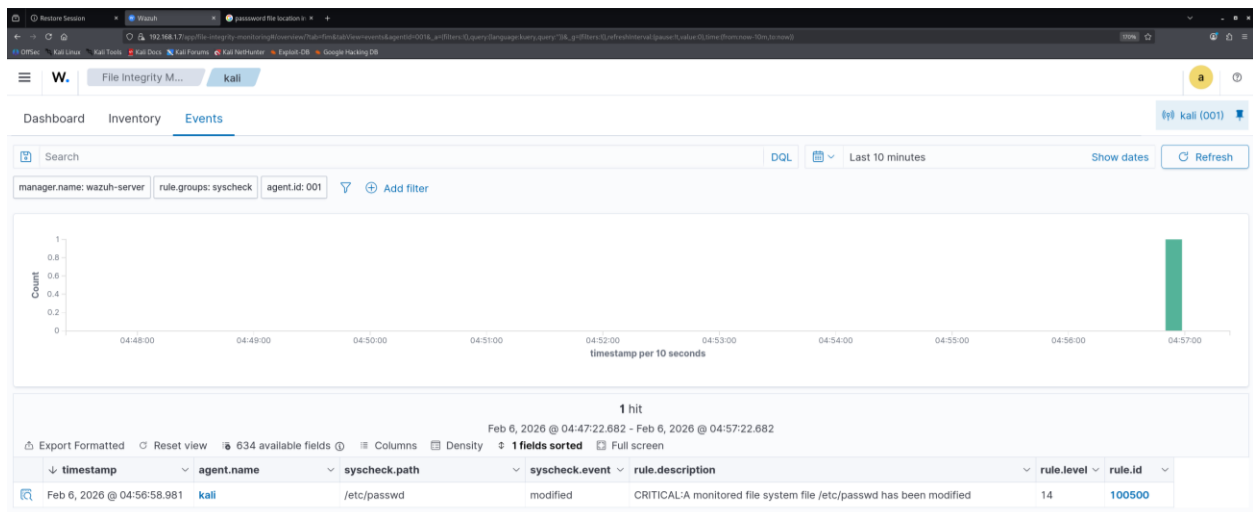
- /etc/passwd file has been accessed and added a few lines through linux terminal

```
Session Actions Edit View Help
# sudo nano /var/ossec/etc/ossec.conf

# sudo systemctl daemon-reload
# sudo systemctl enable wazuh-agent
# sudo systemctl start wazuh-agent

# sudo nano /etc/passwd
```

- Viewing Events in Wazuh Dashboard



The dashboard displayed real-time alert for sensitive file modification

Result:

The custom rules were successfully implemented and verified.

Wazuh provides visibility into:

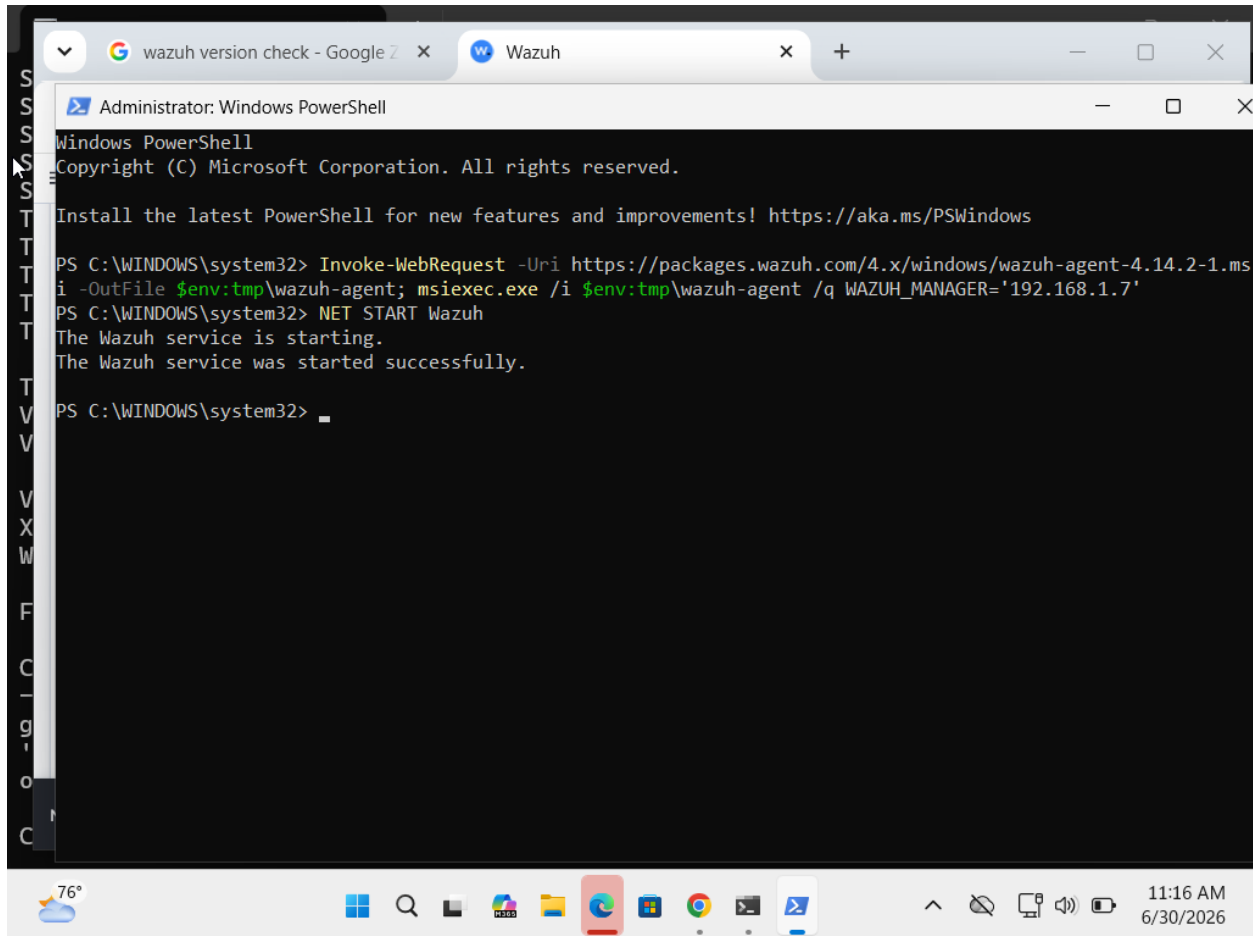
- File integrity
- proactive incident response

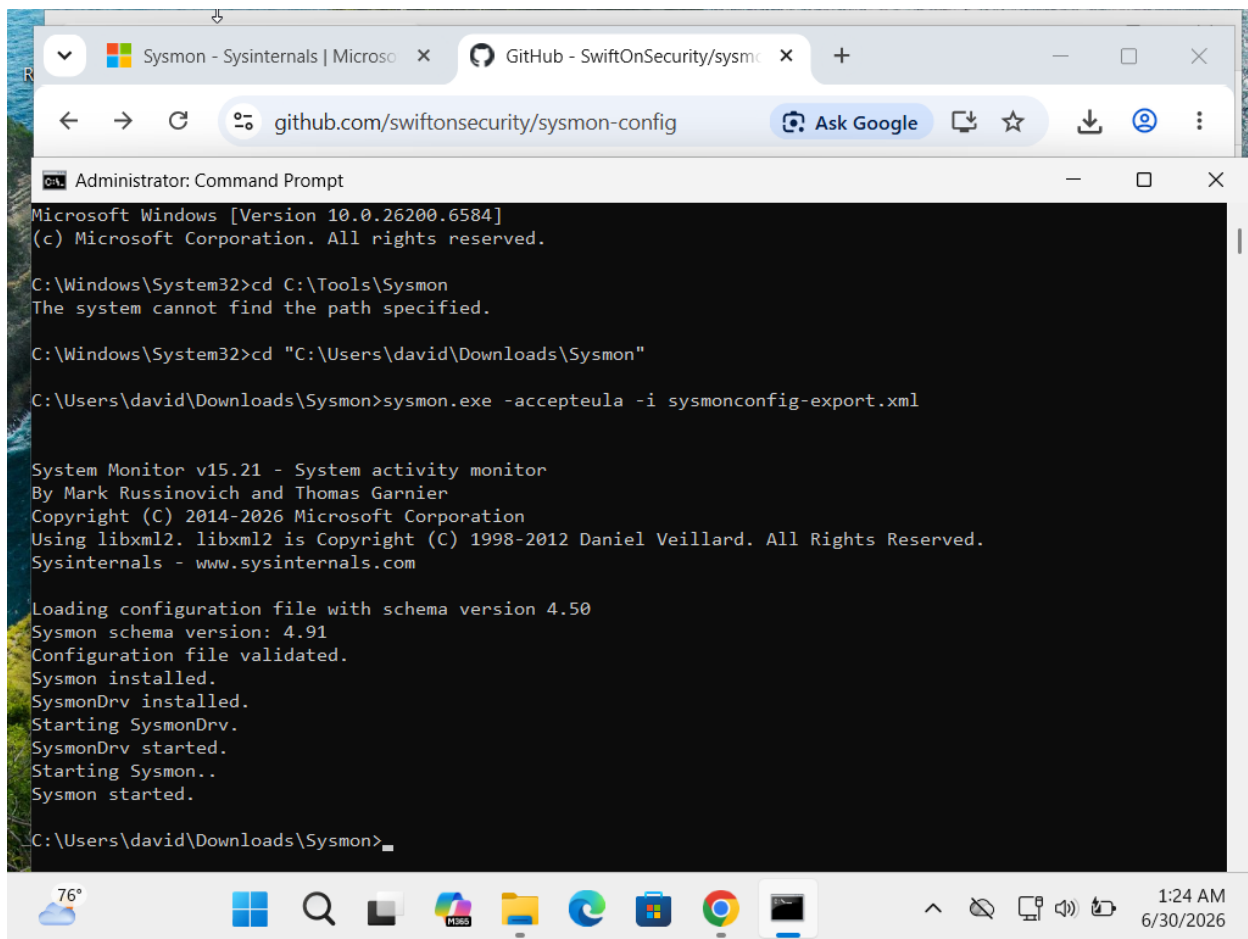
This configuration enhance system security by monitoring file integrity, generating meaningful alerts, and supporting proactive incident response.

8. Installation of Wazuh Agent and Sysmon on Windows 11

Run this command on the Windows PowerShell:

- C:\WINDOWS\system32> Invoke-WebRequest -Uri <https://packages.wazuh.com/4.x/windows/wazuh-agent-4.14.2-1.msi> -OutFile \$env:tmp\wazuh-agent; msixexec.exe /i \$env:tmp\wazuh-agent /q WAZUH_MANAGER='192.168.1.7'
- NET START Wazuh





Type the following command inside “C: \Program Files (x86)\ossec-agent\ossec.conf”:

<localfile>

<location>Microsoft-Windows-Sysmon/Operational</location>

<log_format>eventchannel</log_format>

</localfile>

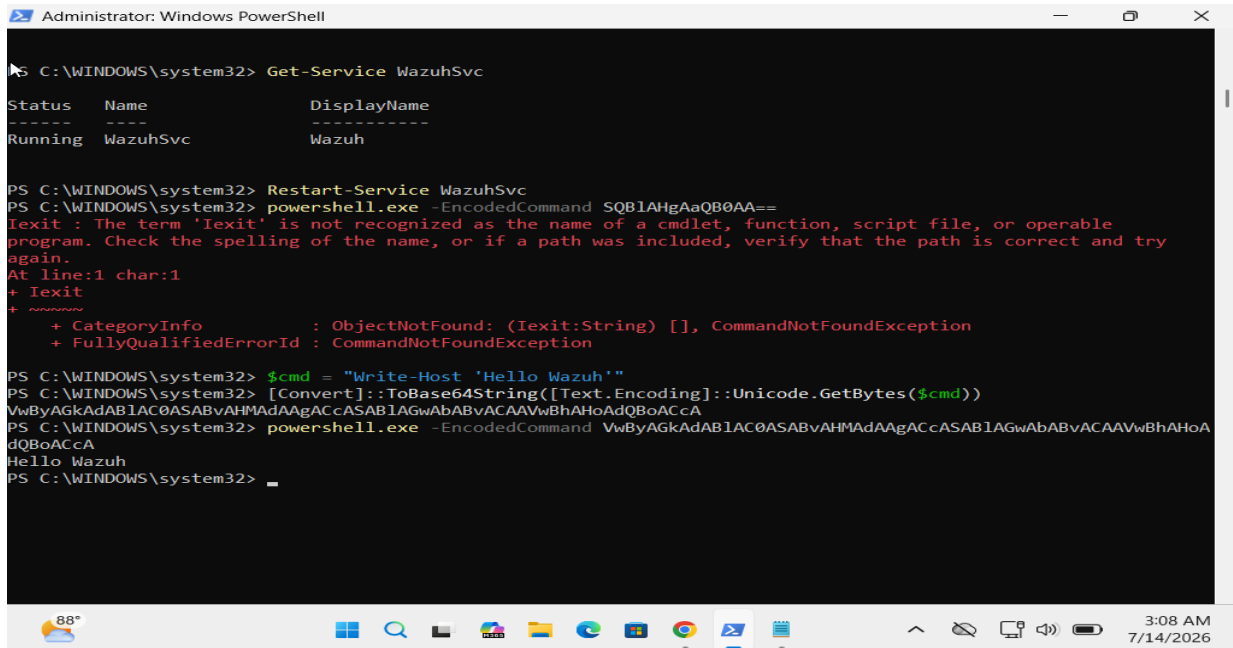
9. Testing Attacks and Verifying Alerts

1. Encoding Command:

Powershell.exe -EncodedCommand

VwByAGkAdABlAC0ASABvAHMAAdAAgACcASABiAGwAbABvACAAVwBhAhoAdQ
BoACcA

Hello Wazuh



```
Administrator: Windows PowerShell

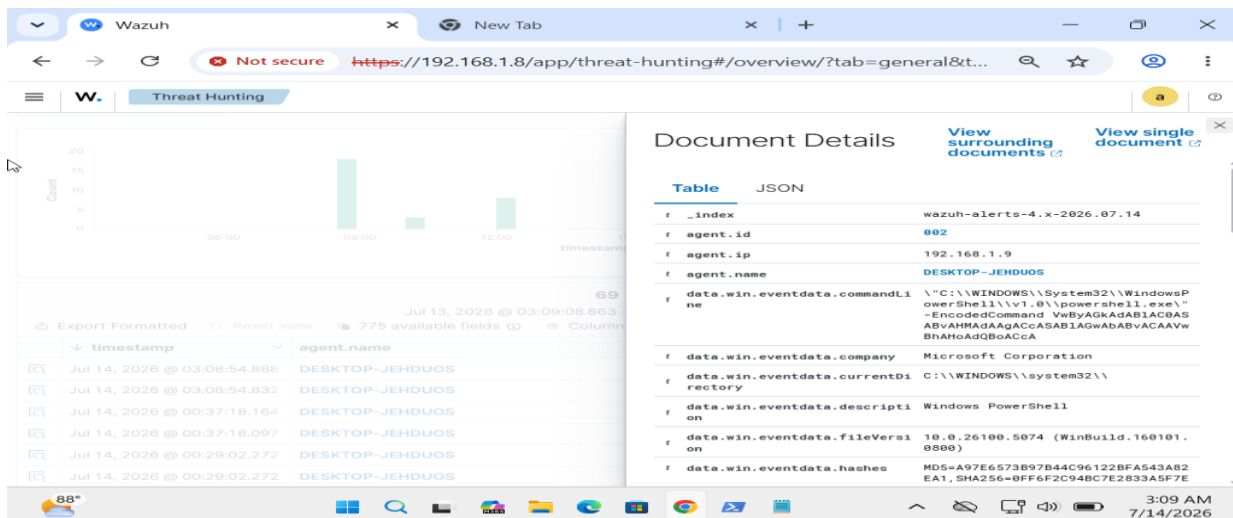
C:\WINDOWS\system32> Get-Service WazuhSvc

Status      Name      DisplayName
-----
Running     WazuhSvc  Wazuh

PS C:\WINDOWS\system32> Restart-Service WazuhSvc
PS C:\WINDOWS\system32> powershell.exe -EncodedCommand SQBlAHgAaQB0AA==
Iexit : The term 'Iexit' is not recognized as the name of a cmdlet, function, script file, or operable
program. Check the spelling of the name, or if a path was included, verify that the path is correct and try
again.
At line:1 char:1
+ Iexit
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Iexit:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\system32> $cmd = "Write-Host 'Hello Wazuh'"
PS C:\WINDOWS\system32> [Convert]::ToBase64String([Text.Encoding]::Unicode.GetBytes($cmd))
VwByAGkAdABlAC0ASABvAHMAAdAAgACcASABiAGwAbABvACAAVwBhAhoAdQBoACcA
PS C:\WINDOWS\system32> powershell.exe -EncodedCommand VwByAGkAdABlAC0ASABvAHMAAdAAgACcASABiAGwAbABvACAAVwBhAhoAdQBoACcA
Hello Wazuh
PS C:\WINDOWS\system32>
```

Monitoring Wazuh Dashboard:



2. Defense Evasion:

Wetutil cl Security

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> Restart-Service -Name wazuh
PS C:\WINDOWS\system32> stop-Service -Name wazuhSvc
PS C:\WINDOWS\system32> Restart-Service WazuhSvc
PS C:\WINDOWS\system32> Restart-Service -Name wazuh
PS C:\WINDOWS\system32> notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
PS C:\WINDOWS\system32> Restart-Service -Name wazuh
PS C:\WINDOWS\system32> wevtutil cl Security
PS C:\WINDOWS\system32>
```

Monitoring Wazuh Dashboard:

The screenshot shows the Wazuh Threat Hunting dashboard in a web browser. The URL is <https://192.168.1.7/app/threat-hunting#/overview/?tab=ge...>. The dashboard has a sidebar with a menu and a main content area. The main content area is divided into two sections: a table of events and a detailed view of a selected event.

Table of Events:

timestamp	agent.name
Jul 16, 2026 @ 23:08:07.508	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:07:55.909	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:07:35.456	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:07:30.483	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:07:24.459	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:07:15.253	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:07:09.341	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:07:09.338	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:07:02.382	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:06:39.953	DESKTOP-JEHUOS
Jul 16, 2026 @ 23:05:01.916	DESKTOP-JEHUOS

Document Details:

data.win.system.level 4

data.win.system.message "The audit log w as cleared.
Subject: Security
ID: S-1-5-21-2641568455-2802-194792-1381794067-1001
Account Name: david
Domain: N
ame: DESKTOP-JEHUOS
Logon I
D: 0x8787B"

3. Monitoring Brute-force Attempt:

Repeated login authentication failure

The screenshot shows the Wazuh Threat Hunting interface. The top bar indicates "14 hits" for the rule "Logon Failure - Unknown user or bad password" (rule.id: 60122). The table below lists the hits with columns: timestamp, agent.name, rule.description, rule.level, and rule.id. The hits are sorted by rule.id.

timestamp	agent.name	rule.description	rule.level	rule.id
Jul 17, 2026 @ 03:21:54.595	DESKTOP-JEHUOS	Sysmon - Suspicious Process	12	61618
Jul 17, 2026 @ 03:21:47.164	DESKTOP-JEHUOS	Logon Failure - Unknown user or bad password	5	60122
Jul 17, 2026 @ 03:21:42.042	DESKTOP-JEHUOS	Logon Failure - Unknown user or bad password	5	60122
Jul 17, 2026 @ 03:21:37.741	DESKTOP-JEHUOS	Logon Failure - Unknown user or bad password	5	60122
Jul 17, 2026 @ 03:21:33.936	DESKTOP-JEHUOS	Logon Failure - Unknown user or bad password	5	60122
Jul 17, 2026 @ 03:21:30.367	DESKTOP-JEHUOS	Logon Failure - Unknown user or bad password	5	60122
Jul 17, 2026 @ 03:21:29.086	DESKTOP-JEHUOS	Powershell process creation	9	92205
Jul 17, 2026 @ 03:21:12.845	DESKTOP-JEHUOS	Sysmon - Suspicious Process	12	61618
Jul 17, 2026 @ 03:19:10.045	DESKTOP-JEHUOS	Software protection service	3	60642
Jul 17, 2026 @ 03:15:32.167	DESKTOP-JEHUOS	Sysmon - Suspicious Process	12	61618
Jul 17, 2026 @ 03:15:12.215	DESKTOP-JEHUOS	Evidence of new service creation	3	92307
Jul 17, 2026 @ 03:14:28.689	DESKTOP-JEHUOS	Sysmon - Suspicious Process	12	61618
Jul 17, 2026 @ 03:14:19.850	DESKTOP-JEHUOS	Executable dropped in Windows	6	92217
Jul 17, 2026 @ 03:13:40.460	DESKTOP-JEHUOS	Executable dropped in Windows	6	92217

The screenshot shows the Wazuh Threat Hunting interface with the "Document Details" panel open. The panel displays the following information:

- data.win.system.version:** 8
- decoder.name:** windows_eventchannel
- id:** 1784283707.770449
- input.type:** log
- location:** EventChannel
- manager.name:** wazuh-server
- rule.description:** Logon Failure - Unknown user or bad password
- rule.firedtimes:** 5
- rule.gdpr:** IV_35.7.d, IV_32.2
- rule.gpg13:** 7.1
- rule.groups:** windows, windows_security, authentication_failed
- rule.hipaa:** 164.312.b
- rule.id:** 60122
- rule.level:** 5
- rule.mail:** false
- rule.mitre.id:** T1531
- rule.mitre.tactic:** Impact
- rule.mitre.technique:** Account Access Removal

4. Discovery Activity:

Command: net user

```
Administrator: Windows PowerShell
Domain: WORKGROUP
Logon Server: \\DESKTOP-JEHUOS
Hotfix(s): 6 Hotfix(s) Installed.
[01]: KB5100998
[02]: KB5054156
[03]: KB5095189
[04]: KB5101650
[05]: KB5094135
[06]: KB5120102
Network Card(s): 1 NIC(s) Installed.
[01]: Intel(R) PRO/1000 MT Desktop Adapter
Connection Name: Ethernet
DHCP Enabled: Yes
DHCP Server: 192.168.1.1
IP address(es)
[01]: 192.168.1.10
[02]: fe80::c005:391:c677:c610
Virtualization-based security: Status: Not enabled
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:
Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.
PS C:\WINDOWS\system32> net user

User accounts for \\DESKTOP-JEHUOS
-----
Administrator          david                      DefaultAccount
Guest                  WDAGUtilityAccount
The command completed successfully.
PS C:\WINDOWS\system32>
```

Monitoring Wazuh Dashboard:

The screenshot shows the Wazuh Threat Hunting dashboard. The left panel displays a list of events with columns for timestamp and agent name. The right panel shows the 'Document Details' for a selected event, including fields like decoder.name, id, input.type, location, manager.name, rule.description, rule.firedtimes, rule.groups, rule.id, rule.level, rule.mail, rule.mitre.id, rule.mitre.tactic, rule.mitre.technique, and timestamp.

timestamp	agent.name
Jul 23, 2026 @ 10:09:31.158	DESKTOP-JEHUOS
Jul 23, 2026 @ 10:09:31.106	DESKTOP-JEHUOS
Jul 23, 2026 @ 09:59:51.328	DESKTOP-JEHUOS
Jul 23, 2026 @ 09:58:29.491	DESKTOP-JEHUOS
Jul 23, 2026 @ 09:52:52.744	DESKTOP-JEHUOS
Jul 23, 2026 @ 09:45:01.010	DESKTOP-JEHUOS
Jul 23, 2026 @ 09:40:05.832	DESKTOP-JEHUOS
Jul 23, 2026 @ 09:39:31.273	DESKTOP-JEHUOS
Jul 23, 2026 @ 09:39:25.979	DESKTOP-JEHUOS
Jul 23, 2026 @ 09:38:16.201	DESKTOP-JEHUOS

Field	Value
decoder.name	windows_eventchannel
id	1784826571.601569
input.type	log
location	EventChannel
manager.name	wazuh-server
rule.description	Discovery activity executed
rule.firedtimes	1
rule.groups	sysmon, sysmon_eid1_detections, windows
rule.id	92031
rule.level	3
rule.mail	false
rule.mitre.id	T1087
rule.mitre.tactic	Discovery
rule.mitre.technique	Account Discovery
timestamp	Jul 23, 2026 @ 10:09:31.158